

# Two's complement

Unsigned numbers only go one way

# Sign bit, negate, range

- At a fixed width, the top bit is the sign
- All ones is minus one
- To negate, invert every bit and add one
- The range is not symmetric
- That asymmetric floor is easy to forget when you “just add one more.”

# Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

## Two's complement lab

Explore **signed range**, **negate** (invert + 1), and **wrap** when values leave the representable window.

**Starter example:** Width 8, pattern for +5 (0000\_0101). Negate to see invert+1  $\rightarrow$  -5. Try min/max and a value that wraps.

Load starter example

Challenges 0 / 22 cleared

Quiz: MSB: In two's complement, the MSB is the? Answer: sign

Answer

Show hintCheckNextIdle

Quiz: MSBQuiz: all onesQuiz: negateQuiz: 8-bit minStarter +5Negate +5Set -1Set minSet max

Wrap 200Width 4 -1Quiz: 8-bit maxInvert +5Toggle MSBNegate min+1 from -1Quiz: 4-bit range

Encode -516-bit -1Quiz: same bits-1 from 0Full story

Core ideas

**Signed range**  
Width  $w$ : from  $-2^{(w-1)}$  to  $2^{(w-1)}-1$ .

**Negate**  
Bitwise invert, then add 1 (mod  $2^w$ ). All ones is -1.

Two's complement starter

# Workbook practice

- In the workbook track, write width eight on paper
- Start from plus five: bits ending in zero one zero one
- Negate by hand, invert, then add one, and confirm you land on minus five
- Then write the signed meanings of all ones, the minimum pattern with only the sign bit set
- Name one RTL pitfall: treating a signed wire as unsigned, or forgetting the asymmetric min

# Pitfalls to watch

- Do not read the most significant bit as “just another magnitude bit” when the type is
- Do not expect plus one hundred twenty-eight to fit in eight-bit two’s complement, it wraps
- And remember: the browser lab is literacy
- Waveforms and RTL still demand the same signed-versus-unsigned discipline

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, negate one value by hand and list min, max, and minus one for one width
- When you are ready, take the short quiz, then continue to overflow and wrap